

AKASH SARAVANAN

Geospatial Analyst & GIS Developer

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PROFILE

Geospatial analyst with a civil-engineering foundation and 4+ years across Earth Observation, remote sensing and GIS — now also building web-based GIS platforms end to end (React/Leaflet frontends, Python backends). Experienced in LiDAR and photogrammetric processing, spectral analysis, cadastral mapping at 200–2,000-acre scale, and Python automation (PyQGIS, Rasterio, GDAL) that removes manual grind from spatial workflows.

EXPERIENCE

GIS Analyst

Sep 2025 — Present

FarmwiseAI · Chennai

- Building a web-based GIS platform for a government transport authority — React/Leaflet map interface over a Python backend, with DWG/DXF-to-web ingestion and 6,000+ interactive parcel features per view.
- Conducting techno-economic feasibility studies (TEFR) and multi-criteria site-suitability models for proposed industrial parks — land potential, connectivity, investment viability.
- Executing end-to-end AutoCAD FMB stitching into high-accuracy industrial layouts; leading QGIS-based QA/QC for property-tax geospatial projects with 100% data-integrity targets.
- Automating daily GIS workflows with Python; applying GeoAI models for predictive land-use analysis and automated feature extraction.

GIS Analyst

Jan 2023 — Aug 2025

Operational Energy Group India (incl. OEG Defence & Aerospace, Zuppa GEN5) · Chennai

- Built GIS datasets, geodatabases and topographic maps for engineering and defence applications in ArcGIS Pro and QGIS.
- Developed ArcGIS Online dashboards for coal-stockpile inventory; performed volumetric analysis and topographic mapping for thermal power plants.
- Planned and flew drone survey missions, processed data (geotagging, photogrammetry) and delivered technical demonstrations to clients and at industry expos.

Site Engineer

Dec 2021 — Dec 2022

SSG Engineers · Salem

- Supervised daily construction against plans and safety standards; coordinated contractors, suppliers and material deliveries; ran workmanship and material quality checks with full site documentation.

SELECTED PROJECTS

Government WebGIS Platform — single-view map review portal: 6,000+ interactive plot features, CAD ingestion with auto-classification, measurement and comparison tools, locked-down stakeholder review mode.

PyQGIS QC Automation, Coimbatore City — automated spatial-error detection between building-footprint and geocode shapefiles; 25% reduction in manual QC workload citywide.

Photogrammetry & Volumetric Estimation — DSMs, orthomosaics and contour maps for National Highway, railway and river-bed corridors; point-cloud volumetrics for coal stockpiles.

Industrial Park Site Suitability & TEFR — multi-criteria decision models and geospatial inputs to feasibility reports guiding investment and regulatory decisions.

Cadastral FMB Stitching at Scale — seamless, legally-defensible parcel maps for sites of 200–2,000 acres across multiple projects.

SKILLS

GIS & EO Tools: QGIS, ArcGIS Pro, Google Earth Engine, ENVI, AutoCAD, ArcGIS Online

Remote Sensing: LiDAR processing, hyperspectral & multispectral analysis, NDVI/EVI/SAVI, image classification, photogrammetry

Python & Automation: Python, PyQGIS, Rasterio, GDAL, GeoPandas, Shapely, NumPy

Web GIS: React, Leaflet, JavaScript, PostGIS, REST APIs

Field & Survey: Drone operations, cadastral/FMB mapping, volumetric analysis, topographic mapping

EDUCATION & CERTIFICATIONS

B.E. Civil Engineering — Panimalar Engineering College, Chennai (CGPA 7.32/10.0) 2020

Diploma in Civil Engineering — Thiagarajar Polytechnic College, Salem (86.74%) 2017

Advanced GIS & Remote Sensing (ArcGIS Pro, QGIS) · AutoCAD Certification